Curs PC.2. 6

Metode divide et impera

① **Exponențiere rapidă**

$a^n \to \Theta(n)$

$$a^n = \begin{cases} 1, & \text{dc } n = 0 \\ a, & \text{dc } n = 1 \\ (a^2)^{\frac{n}{2}}, & \text{dc } n \text{ este par} \\ a \cdot (a^2)^{\frac{n}{2}}, & \text{dc } n \text{ este impar} \end{cases}$$

$O(\log_2 n)$

$fexp(a, n) = a^n$

$fexp(2, 10) = (2^2)^5$

$= fexp(4, 5)$

$= 4 \cdot (4^2)^2$

$= 4\, fexp(16, 2)$

$= 4 \cdot 256$

$= 4 \cdot fexp(256, 1)$

$= 4 \cdot 256 = 1024$

Algoritmul lui Strassen

Algoritmul lui Karatsuba

② **Sortare prin interclasare (Mergesort)**

John von Neumann ~ 1945

$a = (5, 7, 7, 10, 12, 20)$ $m = 6$

$b = (6, 7, 25, 30)$ $n = 4$

$r = (5, 6, 7, 7, 7, 10, 12, 20, 25, 30)$ $m + n = 10$

interclasare $\longrightarrow O(m + n)$

t:	5	7	10	12	3	4	10
r	3	4	5	7	10	10	12

$O(dr - k + 1) \leq O(n)$

```
void mergesort (int v[], int st, int dr)
{ int mij;
    if (st < dr) {
        mij = (st + dr)/2;
        mergesort (v, st, mij);
        mergesort (v, mij+1, dr);
        interclasare (v, st, mij, dr);
    }
}
```

$$T(n) = \begin{cases} 1, \text{ dc } n = 1 \\ 1 + 2 \cdot T\left(\frac{n}{2}\right) + n, \ n \geq 2 \end{cases}$$

Pt $n = 2^k \Rightarrow T(n) = T(2^k) = 2 \cdot T(2^{k-1}) + 2^k =$

$= 2\left[2T(2^{k-2}) + 2^{k-1}\right] + 2^k = 2^2 \cdot T(2^{k-2}) + 2^k + 2^k =$

$= 2^k \cdot \underbrace{T(2^0)}_{1} + \sum_{i=1}^{k} 2^k$

$= \sum_{i=1}^{k+1} 2^k = n \cdot \log_2 n + 1$

③ Soltde rapidă (Quicksort)

C.A.R. Hoare ~ 1960

Tablou sortat descr ⟺ dice elem este pivot

$v = (\underbrace{7, 5, 10, 4,}_{\leq}\ \boxed{14}\ \underbrace{20, 17, 15}_{\geq})$

PIVOT
rămâne pe ultima poz. în tabelul sortat desc

$v = (7, 5, 10, 4, 14, 20, 17, 15)$ Partiționarea Nico Lomuto (1987)

st=0 ; dr=7 ; pivot = 15

$v = (\boxed{7}, 5, \cancel{10}\ 4, \cancel{4}\ 10, \cancel{14}\ 14, 20, 17, \boxed{15})$

st 1 ← i = -1 ; j = 0 ;

poz. ult. elem ≤ pivot

⇒ $v = (7, 5, 4, 14, 15, 20, 17, 10)$

```
void quicksort(int v[], int st, int dr)
{ int poz_pivot;
  if (st < dr){
    poz_pivot = partitionare(v, st, dr)
    quicksort(v, st, poz_pivot - 1);
    quicksort(v, poz_pivot, dr);
  }
}
```

```
int partitionare(int v[], int st, int dr)
{ int i, j, pivot = v[dr]
  i = st-1;
  for(j = st; j < dr; j++){
    if(v[j] <= pivot){
      i++;
      interschimbare(&v[i], &v[j]);
    }
  interschimbare(&v[i+1], &v[dr]);
  return i+1;
}
```

$\alpha c \; k \approx \frac{n}{2} \Rightarrow T(n) = 2T\left(\frac{n}{2}\right) + n$ $\qquad T(n) = \begin{cases} n + T(k) + T(n-k), & n \geq 2 \\ 1, & n = 1 \end{cases}$

$\Rightarrow O(n \log_2 n)$

$\alpha c \; k \approx 1 \Rightarrow T(n) = n + T(n-1)$